

Small-polaron transport

11.1 Loss of coherence and hopping transport

Incoherent transport results when the energy emitted or absorbed in moving a carrier exceeds the energy characterizing iso-energetic transfer. This section illustrates why small-polaron motion is usually incoherent. In summary, coherence is lost at high temperatures where atoms move classically when the change in the energy of a small-polaron's self-trapped carrier as it moves between sites is greater than its electronic transfer energy. Coherence is lost at low temperatures when a small-polaron's transfer energy is so reduced by the requirement that atoms' quantum-mechanically tunnel that it is overwhelmed by site-to-site variations of a small-polaron's energy.

Consider a small polaron formed with the Holstein short-range electron–phonon interaction. Then the requirement for a small-polaron's energetic stability in a cubic crystal becomes $F^2/2k > W/2$, where, as earlier, F represents the force associated with the short-range electron–phonon interaction, k is the displaced atoms' stiffness constant and W is the electronic bandwidth [cf. Eq. (2.5) and Section 4.4]. In particular, the carrier-induced shift of atoms' equilibrium positions by F/k lowers the self-trapped carrier's energy by $F(F/k) = F^2/k$ while increasing the atomic strain energy by $k(F/k)^2/2 = F^2/2k$.

For a self-trapped carrier to move the associated atoms must alter their positions. With classical atomic movement, the self-trapped carrier's energy must be brought into coincidence with what it would be upon its occupying a different site. The minimum-energy coincidence raises the electronic energy at the occupied site by $F(F/2k) = F^2/2k$, halfway toward liberating the self-trapped carrier, while creating an equivalent potential well at the carrier's would-be destination. Such small-polaron motion is incoherent when the increase of the electronic energy exceeds this motion's coherence energy t , the electronic transfer energy: $F^2/2k > t$. This condition is generally fulfilled because stable small-polaron formation

insures that $F^2/2k > W/2 = zt$, where z is the number of nearest-neighbors in the tight-binding approximation.

The periodicity of an ideal lattice guarantees that a small polaron might still move coherently at low temperatures where atoms' classical motions are suppressed. Then a small-polaron's self-trapped carrier moves between sites as the associated atoms tunnel so as to remove atomic displacements about an initially occupied site and create equivalent atomic displacements about the destination site. The bandwidth associated with such iso-energetic motion is greatest in the adiabatic limit where an electronic carrier is always able to follow atomic movement. Then, as indicated previously [cf. Eq. (4.42)], the zero-temperature adiabatic small-polaron bandwidth is

$$W_{\text{SP}} = 2z\hbar\omega \sqrt{\left(\frac{F^2/2k}{\hbar\omega}\right)} \exp\left(-\frac{F^2/2k}{\hbar\omega}\right). \quad (11.1)$$

This bandwidth vanishes in the classical limit $\hbar \rightarrow 0$ where atomic tunneling is proscribed. Furthermore, as described in Section 10.2, bandwidths arising from atomic tunneling of harmonically vibrating atoms fall with increasing temperature. Thus, the small-polaron band is widest for adiabatic motion at zero temperature. Nonetheless, the small-polaron bandwidth of Eq. (10.1) is extremely small. For example, with $(F^2/2k)/\hbar\omega = 10$ the small-polaron bandwidth is about three orders of magnitude smaller than the displaced atoms' characteristic vibration energy $\hbar\omega$.

The small-polaron bandwidth, the coherence energy for this motion, is generally so small that it is less than the disorder potential resulting from defects and impurities. Indeed, even the potential drop between sites separated by $10 \text{ \AA}$ induced by an applied field of 10^4 V/cm , as applied in transit experiments, exceeds the small-polaron bandwidth. The applied field is itself then enough to obliterate coherent motion. The coherent motion of small bipolarons is even easier to destroy since having two carriers move together has the effect of replacing F by $2F$ in Eq. (11.1). Thus coherent small-polaronic motion is generally suppressed. As a result small polarons and small bipolarons are usually envisioned as moving incoherently by sequences of thermally assisted hops.

11.2 Two complementary types of hopping transport

Phonon-assisted electronic hopping is exemplified (1) by carriers' low-temperature hopping between shallow impurities in conventional covalent semiconductors and (2) by carriers' high-temperature hopping between cations in some transition-metal oxides. In both cases the hopping rates rise with increasing temperature. However,

there are qualitative differences between these two examples. These distinctions define two complementary types of hopping transport: shallow-impurity hopping and small-polaron hopping.

Phonon-assisted hopping between localized electronic states is driven by their electron–phonon interactions. These interactions facilitate the absorption and emission of phonons that accompany a phonon-assisted jump. Electron–phonon interactions are most effective for phonons whose wavelengths exceed localized electronic-states' radii.

Since the radii of the electronic states associated with shallow impurities in conventional semiconductors R_i are much larger than the interatomic separation a , $R_i \gg a$, these states only interact with the small fraction of phonons whose wavelengths are exceptionally long. As a result large-radius electronic states have a weak net interaction with phonons [e.g. $(F^2/2k)(a/R_i)^3/\hbar \omega < 1$]. Each hop between such states is then dominated by the absorption or emission of just a single long-wavelength phonon (Miller and Abrahams, 1960; Emin, 1974; 1975a). The rate for a thermally assisted jump in which a single phonon is absorbed is then proportional to the Bose factor $1/[\exp(\Delta/\kappa T) - 1]$, where Δ , the energy of the absorbed phonon, equals the energy difference between the hop's two electronic states. For an acoustic-phonon-assisted jump the restriction of the electron–phonon interaction to long-wavelength phonons limits Δ to a small fraction of the Debye energy. Thus the low-temperature jump rate for a hop upward in energy is thermally activated with an activation energy that is much smaller than the Debye energy. Indeed, typical activation energies for shallow-impurity conduction are about 10^{-4} eV– 10^{-3} eV (Miller and Abrahams, 1960). The Bose factor ensures that this jump rate will rise in proportion to the temperature for $\kappa T \gg \Delta$.

A hop downward in energy between large-radii shallow-impurity states primarily occurs with emission of only a single very low-energy acoustic phonon. Downward jumps between sites having a greater energy disparity are thereby precluded. Negative differential conductivity can even result if hopping transport is impeded by the inter-site potential drop generated by an applied electric field exceeding the highest energy acoustic phonon with which a shallow impurity can interact (Emin and Hart, 1985).

Small-polaron hopping refers to the phonon-assisted jumping of a carrier through severely localized (collapsed) states. The severe localization of these states enables them to interact strongly with phonons of nearly all wavelengths thereby yielding a strong net electron–phonon interaction [e.g. $(F^2/2k)/\hbar \omega \gg 1$]. As a result, a small-polaron jump usually involves the absorption and emission of multiple phonons. Indeed, even at low temperatures small-polaron hops typically involve far more than the minimal number of phonons required for energy conservation (Emin, 1974). As the temperature is raised above the characteristic phonon temperature the jump rate

risks in an Arrhenius manner with an activation energy that generally exceeds the characteristic phonon energy. This high-temperature behavior distinguishes small-polaron hopping from that between electronic states which are weakly coupled to phonons.

Hopping-type impurity conduction in conventional semiconductors usually involves very long hops between large-radius states. In particular, hopping-type impurity conduction is reported at dopant concentrations below $2 \times 10^{17} \text{ cm}^{-3}$ in *n*-type silicon and below $6 \times 10^{15} \text{ cm}^{-3}$ in *n*-type germanium (Miller and Abrahams, 1960). The spherically averaged radii of ellipsoidal donor-state wavefunctions in *n*-type silicon and germanium are about 20 Å and 45 Å, respectively. Thus, the inter-impurity separations are an order of magnitude larger than the donor-state radii. In these situations the transfer energy linking the impurity centers is envisioned as the small perturbation that enables a hop to take place. In other words, these hops are viewed as *non-adiabatic*.

A different situation often prevails for small-polaron hopping in transition-metal oxides. In particular, the distances between nearest-neighbor transition-metal ions are just several times these cation radii. In these circumstances the electronic transfer energies can be too large to treat them simply as perturbations (Herring, 1961). The small-polaron jumps are then to be treated as *adiabatic* (Herring, 1961; Holstein 1959b; Emin and Holstein, 1969; Emin, 2008).

11.3 Overview of polaron hopping

A small-polaron hop may be regarded as a three-step process. A jump begins when, amidst atoms' thermal vibrations, extraordinarily large atomic displacements occur near a self-trapped carrier. Appropriate large-amplitude fluctuations enable the self-trapped carrier to then transfer between sites. Finally these transitory large-amplitude atomic displacements relax, dissipating energy to the vibrations of surrounding atoms.

Exchange of vibration energy between atoms, governed by the vibrations' dispersion, is crucial to phonon-assisted hopping. Effects of vibrations' dispersion on small-polaron hopping will be explicitly considered in Sections 11.4 through 11.7. However this section's overview of the physics of a small-polaron jump only implicitly presumes the presence of adequate vibrational dispersion.

Figure 11.1 illustrates the evolution of the processes which dominate the transfer of a self-trapped carrier as the temperature is increased (Emin, 1982). At low temperatures hopping is dominated by transfer processes that cost little in strain energy as atoms remain close to their equilibrium positions. In particular, process *a* of Fig. 11.1 depicts a self-trapped carrier being transferred as atoms tunnel from positions commensurate with its occupying its initial site to equivalent positions that

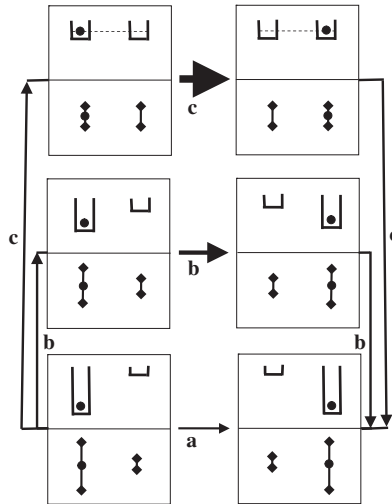


Fig. 11.1 A small-polaron hop occurs as a self-trapped carrier (denoted with a black dot) moves from a potential well centered on one site to an equivalent potential well centered on another (top subpanels) in response to changes of the atomic displacements at the two sites (bottom subpanels). The process requiring the smallest strain energy (process *a*) has the atomic displacements at the occupied site (schematically illustrated by a vertical line linking two diamonds) tunneling to values corresponding to the unoccupied site, while the atomic displacements at the initially unoccupied site tunnel to those for the occupied site. The speed of the transfer is indicated by the width of the central arrow which denotes it. Faster transfer occurs (in process *b*) when strain energy is first expended to alter the disparity of the configurations between which atoms tunnel. As depicted by process *c*, expending sufficient strain energy can eliminate the need for atomic tunneling between disparate atomic displacement patterns. The electronic carrier then transfers relatively rapidly between electronic states that are temporarily coincident.

correspond to the carrier being self-trapped at its final site. This atomic tunneling slows the transfer process. Thus, as indicated by arrow *a* in Fig. 11.2, the predominance of such atomic tunneling processes at low temperatures produces a relatively small jump rate.

Raising the temperature facilitates atoms assuming configurations requiring greater strain energy. As illustrated by process *b* in Fig. 11.1, atoms can then depart further from their equilibrium locations to positions that enable them to tunnel between less disparate configurations. Atomic tunneling's slowing of the transfer process is thereby reduced. As a result, depicted by arrow *b* in Fig. 11.2, the small-polaron hopping rate rises as the temperature is raised.

At high enough temperatures a small-polaron hop is dominated by transfers in which atoms surrounding initial and final sites are displaced sufficiently to bring the

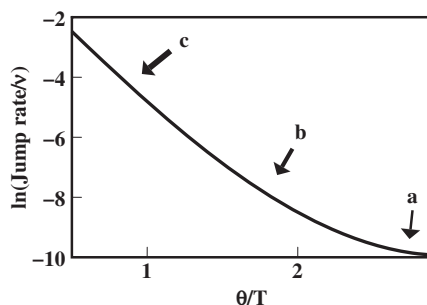


Fig. 11.2 A typical small-polaron jump rate in units of the characteristic atomic vibration frequency ν is plotted against reciprocal temperature in units of the vibration temperature $\theta \equiv \hbar\nu/k$. At low temperatures the rate is very small and increases slowly with rising temperature as low-energy atomic-tunneling processes (e.g. process *a* of Fig. 11.1) predominate. The rate and its temperature dependence both increase with rising temperature as more efficient tunneling processes requiring larger strain energies (e.g. process *b* of Fig. 11.1) manifest themselves. At higher temperatures the rate becomes Arrhenius when transfer processes that eschew atomic tunneling but require relatively large strain energies prevail.

electronic energy of a carrier at the initial site into coincidence with what it would be if it occupied the final site. Then, as depicted by process *c* of Fig. 11.1, the carrier can move between sites without any atomic tunneling. The corresponding jump rate, indicated by arrow *c* in Fig. 11.2, then assumes an Arrhenius temperature dependence with its activation energy being the minimum strain energy required to form a coincidence. Since this activation energy only involves atoms' classical motion, it is independent of $\hbar$.

All told, Fig. 11.1 depicts increasingly energetic strained configurations enabling increasingly rapid electronic transfers. Electronic transfer is facilitated by reducing the associated atomic tunneling. The small-polaron hopping rate of Fig. 11.2 rises with increasing temperature as the probability of obtaining such energetic atomic configurations increases. The rate is non-Arrhenius at temperatures below that characterizing the energy of the phonons with which the self-trapped carrier interacts. At higher temperatures the rate becomes Arrhenius as atomic tunneling is supplanted by atoms' classical motion.

The ability of the carrier to move between sites in response to atomic motions depends on its electronic transfer energy. These electronic transfer energies are generally functions of atomic displacements. Indeed, the potential energies and the electronic wavefunctions that determine these electronic transfer integrals all depend on atoms' positions. Thus, an electronic transfer energy in a rigid solid differs from that governing a self-trapped carrier's optical transitions and from that for its semiclassical hopping (e.g. the electronic transfer energy at a coincidence

event). The dependence of a diffusing particle's transfer energy on atomic deformations can even be so strong that it qualitatively affects hopping diffusion (cf. light-interstitial diffusion in metals, Chapter 17). Nonetheless, the electronic transfer energy for a hop between sites $\mathbf{g}$ and $\mathbf{g} + \mathbf{h}$ is often approximated as the constant one would obtain without polaron formation and any atomic displacements. In a crystal this electronic transfer energy is just the Fourier transform of the carrier's energy as a function of its quasi-momentum $E(\mathbf{k})$:

$$t(\mathbf{h}) = \sum_{\mathbf{k}} E(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{h}}. \quad (11.2)$$

This electronic transfer energy generally falls with increasing inter-site separation $\mathbf{h}$ in a manner that reflects the complexity of the carrier's dispersion relation $E(\mathbf{k})$. Nonetheless, transfer energies are often presumed to simply fall exponentially with jump distance.

The simplest approach to calculating the small-polaron jump rate is to assume that the electronic transfer energy is the small perturbation which enables a carrier to move between sites. Section 11.4 describes a version of this (non-adiabatic) approach that is generalized to address jumps between states that are inequivalent in their energies and local electronic wavefunctions. Form factors permit these states to have finite sizes and non-spherical shapes whereas simpler calculations implicitly approximate the localized electronic states as just stationary points. The calculation yields the full temperature-dependence of the non-adiabatic jump rate. In Section 11.5 the high-temperature semiclassical limit of the non-adiabatic jump rate is also derived using Holstein's occurrence-probability approach. However, the non-adiabatic approach is only applicable if the electronic transfer energy is so small that the carrier rarely transfers during a coincidence event. By contrast, the high-temperature adiabatic regime discussed in Section 11.6 envisions the electronic transfer being large enough (typically $t > \hbar \omega$) so that a carrier always hops at a coincidence event. Two types of adiabatic jumps are discussed. In one scheme coincidences are assumed to develop and relax rapidly (in less than a vibration period). The second scheme envisions coincidences developing and relaxing slowly as associated atomic vibrations possessing comparable frequencies are modulated by inter-oscillator energy transfer. Section 11.7 describes how such weak vibrational dispersion fosters temporal and spatial correlations between coincidence events. These correlations cause a carrier's hops to occur in flurries separated by quiescent periods. The semiclassical approach is then extended in Section 11.8 to address the semiclassical hopping of a singlet bipolaron. It is shown that a singlet bipolaron's two electronic carriers do not generally jump together as a pair. Rather, singlet-bipolarons' semiclassical hopping is dominated by their thermally assisted

decomposition into separated polarons that hop independently of one another. Finally, Section 11.9 describes how polaron hopping in magnetic semiconductors is affected by their exchange interactions. In particular, ordering of a magnetic semiconductor's local moments alters its polarons' hopping conduction.

11.4 Non-adiabatic polaron jump rate

Non-adiabatic small-polaron hopping presumes very small electronic-transfer energies. Thus the self-trapped carrier usually remains localized and only occasionally avails itself of the opportunities to hop which atomic vibrations present to it. In this scheme the vibrations are approximated as just those about a carrier that is confined to a single site. For example, Fig. 11.3 plots that portion of the non-adiabatic (small- t) limit of the vibratory potential for two sites of Holstein's molecular crystal which depends on the two sites' relative displacement parameter $x_2 - x_1$:

$$V_r(x_2 - x_1) \equiv k(x_2 - x_1)^2/4 \pm F(x_2 - x_1)/2, \quad (11.3)$$

where the $\pm$ sign pertains when sites 1 and 2 are, respectively, occupied. Here x_n is the displacement parameter for the n -th molecule. When this molecule is occupied by a carrier its equilibrium displacement is shifted by F/k . The darker curve is the relative potential when site 1 is occupied. Similarly the lighter curve is the relative potential when site 2 is occupied. A non-adiabatic hop from site 1 to site 2

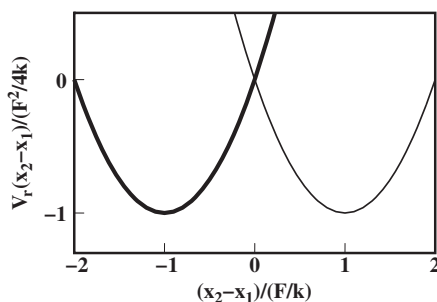


Fig. 11.3 This configuration diagram is often used to describe non-adiabatic hopping between a pair of sites. That portion of the deformational potential energy in the weak-dispersion limit of Holstein's molecular-crystal model, measured in units of $F^2/4k$, which depends on the relative atomic displacement coordinate of two adjacent sites $x_2 - x_1$ is plotted against this parameter, measured in units of F/k . The dark and light curves correspond to the carrier's occupations of sites 1 and 2, respectively. The energy of these curves' crossing point exceeds that of either curves' minimum by $F^2/4k$, the activation energy for non-adiabatic hopping in its high-temperature Arrhenius regime.

occasionally occurs when vibrations take the system through the crossing point at $x_1 = x_2$ or when atomic tunneling transfers the system through the potential barrier separating the two potential wells. Passage through the crossing point corresponds to process *c* of Figs. 11.1 and 11.2. Tunneling through the barrier represents processes *a* and *b* of Figs. 11.1 and 11.2.

The non-adiabatic hopping rate depends on the velocity with which an electronic carrier transfers between sites in response to atoms' movement. Since the set of localized electronic states between which a carrier hops is generally incomplete, the carrier's velocity operator is best expressed as $\mathbf{v} \equiv i(H_e \mathbf{r} - \mathbf{r} H_e)/\hbar$, where H_e is the Hamiltonian describing the carrier's movement (Holstein and Friedman, 1968). This approach insures that the expectation value of the velocity equals the time derivative of the expectation value of the carrier's position. Diagonal matrix elements of a carrier's position between localized electronic states define its location and are unbounded: $\langle \phi_g | \mathbf{r} | \phi_g \rangle = \mathbf{g}$, where $\mathbf{g}$ denotes the centroid of a localized electronic function. By contrast, off-diagonal matrix elements of a carrier's position are bounded: $\langle \phi_{g'} | \mathbf{r} | \phi_g \rangle \rightarrow 0$ as $|\mathbf{g}' - \mathbf{g}| \rightarrow \infty$. DC conduction's unbounded motion is described by a velocity operator that retains only diagonal matrix elements of the position operator: $\langle \phi_{g'} | \mathbf{v} | \phi_g \rangle = i t_{g',g} (\mathbf{g} - \mathbf{g}')/\hbar$, where the transfer energy is defined by $t_{g',g} \equiv \langle \phi_{g'} | H_e | \phi_g \rangle$ (Holstein and Friedman, 1968). However, ac conduction from bound carriers' polarization currents generally requires inclusion of the non-diagonal matrix elements of the position operator. Even a crystal's regular array of potential wells produces bounded polarization-type contributions to the matrix elements of a carrier's position vector (Karplus and Luttinger, 1954). Nonetheless, these polarization contributions do not affect the dc transport phenomenon of Wannier–Stark localization (Anderson, 1963; Emin and Hart, 1987a).

Most generally, a phonon-assisted hop occurs when an electronic carrier moves between sites in response to atoms' vibrations. In the non-adiabatic regime the electronic velocity associated with transferring an electronic carrier between sites is presumed to be very small. Equivalently the electronic transfer energy $t_{g,g'}$ linking localized electronic states centered on sites $\mathbf{g}$ and $\mathbf{g}'$ is treated as the perturbation that enables the transition. The validity of this approach relies on the smallness of the electronic transfer energy.

Calculations of the rate for a non-adiabatic phonon-assisted transition commonly employ the simplification of ignoring carrier-induced shifts of atomic vibrations' frequencies. Then one of several equivalent expressions for this phonon-assisted jump rate becomes

$$R_{g,g'} = \left(\frac{t_{g,g'}}{\hbar} \right)^2 e^{-\Delta_{g,g'}/2kT} e^{-2S_{g,g'}} \int_{-\infty}^{\infty} d\tau e^{F_{g,g'}(\tau)} \cos(\Delta_{g,g'} \tau / \hbar), \quad (11.4)$$

where $\Delta_{g,g'} \equiv E_{g'} - E_g$ is the non-zero difference of the energies of the hop's initial and final states (Yamashita and Kurosawa, 1958; Holstein, 1959b; Emin, 1974; 1975a; Bergeron and Emin, 1977). The other terms in this jump-rate expression are defined by:

$$2S_{g,g'} \equiv N^{-1} \sum_{\mathbf{q}} \Gamma_{g,g'}(\mathbf{q}) \gamma(\mathbf{q}) \coth(\hbar\omega_{\mathbf{q}}/2\kappa T) \quad (11.5)$$

and

$$F_{g,g'}(\tau) \equiv N^{-1} \sum_{\mathbf{q}} \Gamma_{g,g'}(\mathbf{q}) \gamma(\mathbf{q}) \csc h(\hbar\omega_{\mathbf{q}}/2\kappa T) \cos(\omega_{\mathbf{q}}\tau), \quad (11.6)$$

where atoms are presumed to vibrate harmonically among N modes of frequency $\omega_{\mathbf{q}}$ and wavevector $\mathbf{q}$. The spatial correlation factor

$$\Gamma_{g,g'}(\mathbf{q}) \equiv f_g^2 + f_{g'}^2 - 2f_g f_{g'} \cos[\mathbf{q} \cdot (\mathbf{g} - \mathbf{g}')] \quad (11.7)$$

indicates that vibrations whose wavelengths exceed the inter-site separation tend to be ineffective in producing a hop since they do not alter the energy difference between the two sites. Moreover, the form factor

$$f_g(\mathbf{q}) \equiv \langle \phi_g | e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{g})} | \phi_g \rangle \quad (11.8)$$

indicates that a localized electronic state interacts most effectively with phonons whose wavelengths exceed its spatial extent. For example, the form factor for a hydrogenic wavefunction of radius b_g centered at site $\mathbf{g}$ is $f_g(\mathbf{q}) = [1 + (qb_g/2)^2]^{-2}$. Finally, $\gamma(\mathbf{q})$ describes the carrier's electron-phonon interaction; in particular, for the Fröhlich long-range electron-phonon interaction:

$$\gamma_{\text{LR}}(\mathbf{q}) \equiv \frac{e^2 \left(\frac{1}{\epsilon_{\infty}} - \frac{1}{\epsilon_0} \right) \left(\frac{q_D}{q} \right)^2}{\hbar\omega_{\mathbf{q}}}, \quad (11.9)$$

where q_D is the Debye wavevector. The deformation-potential and molecular-crystal-model short-range electron-phonon interactions give:

$$\gamma_{\text{DP}}(\mathbf{q}) \equiv \frac{(Zq_D)^2/2k}{\hbar\omega_{\mathbf{q}}} \quad (11.10)$$

and

$$\gamma_{\text{MCM}}(\mathbf{q}) \equiv \frac{F^2/2k}{\hbar\omega_{\mathbf{q}}}, \quad (11.11)$$

respectively, where Z is the deformation potential energy. All told, the $\mathbf{q}$ -dependences of the arguments of the summations of Eqs. (11.5) and (11.6) are governed by: (1) the nature of the electron–phonon interaction, (2) the sizes and separation of the two electronic states involved in the hop, and (3) the dispersion relations of the involved phonons.

At high enough temperatures with sufficiently strong electron–phonon coupling and dispersion of the vibration frequencies, $F_{\mathbf{g},\mathbf{g}'}(\tau)$ develops a large primary peak at $\tau = 0$ (Holstein, 1959b). The contribution from near this primary peak, rather than the sum of contributions from subsidiary peaks, will then dominate the integral of Eq. (11.4) if the vibration dispersion is three-dimensional (de Wit, 1968; Emin, 1975a).

The contribution to the τ -integration of Eq. (11.4) from the primary peak occurs for $\omega_{\mathbf{q}}\tau \ll 1$. One may therefore write $F_{\mathbf{g},\mathbf{g}'}(\tau) \approx F_{\mathbf{g},\mathbf{g}'}(0) - G_{\mathbf{g},\mathbf{g}'}\tau^2/2$ within the integrand with

$$G_{\mathbf{g},\mathbf{g}'}(T) \equiv N^{-1} \sum_{\mathbf{q}} \Gamma_{\mathbf{g},\mathbf{g}'}(\mathbf{q}) \gamma(\mathbf{q}) \csc h(\hbar\omega_{\mathbf{q}}/2\kappa T) \omega_{\mathbf{q}}^2. \quad (11.12)$$

Upon performing the τ -integration the non-adiabatic small-polaron jump rate becomes

$$R_{\mathbf{g},\mathbf{g}'} = \left(\frac{t_{\mathbf{g},\mathbf{g}'}}{\hbar} \right)^2 \sqrt{\frac{2\pi}{G_{\mathbf{g},\mathbf{g}'}}} e^{-\Delta_{\mathbf{g},\mathbf{g}'}/2\kappa T} e^{-E_{\mathbf{g},\mathbf{g}'}(T)/\kappa T} e^{-(\Delta_{\mathbf{g},\mathbf{g}'}/\hbar)^2/2G_{\mathbf{g},\mathbf{g}'}(T)}, \quad (11.13)$$

where

$$E_{\mathbf{g},\mathbf{g}'}(T)/\kappa T \equiv 2S_{\mathbf{g},\mathbf{g}'} - F_{\mathbf{g},\mathbf{g}'}(0) = N^{-1} \sum_{\mathbf{q}} \Gamma_{\mathbf{g},\mathbf{g}'}(\mathbf{q}) \gamma(\mathbf{q}) \tanh(\hbar\omega_{\mathbf{q}}/4\kappa T). \quad (11.14)$$

As illustrated in Fig. 11.2, this jump rate's temperature dependence becomes Arrhenius at sufficiently high temperatures. Then $E_{\mathbf{g},\mathbf{g}'}(T)$ approaches the temperature-independent value

$$E_{\text{A}}^{\mathbf{g},\mathbf{g}'} \equiv N^{-1} \sum_{\mathbf{q}} [\gamma(\mathbf{q}) \hbar\omega_{\mathbf{q}}] \Gamma_{\mathbf{g},\mathbf{g}'}(\mathbf{q})/4 \quad (11.15)$$

with

$$G_{\mathbf{g},\mathbf{g}'}(T) \rightarrow 8\kappa T E_{\text{A}}^{\mathbf{g},\mathbf{g}'} / \hbar^2. \quad (11.16)$$

As a result the semiclassical limit for the non-adiabatic jump rate becomes:

$$R_{g,g'} = \left(\frac{t_{g,g'}^2}{\hbar} \right) \sqrt{\frac{\pi}{4\kappa T E_{g,g'}}} e^{-(4E_{g,g'} + \Delta_{g,g'})^2 / 16E_{g,g'} \kappa T}. \quad (11.17)$$

The semiclassical jump rate's activation energy generally rises with increasing jump distance. In particular, the dependence of the activation energy of the non-adiabatic semiclassical jump rate in Eq. (11.15) on jump distance enters through the correlation factor Eq. (11.7). At sufficiently large jump distances the activation energy of the non-adiabatic semiclassical jump rate approaches its maximum value:

$$E_A^{\max} \equiv N^{-1} \sum_{\mathbf{q}} [\gamma(\mathbf{q}) \hbar \omega_{\mathbf{q}}] \{f_g^2(\mathbf{q}) + f_{g'}^2(\mathbf{q})\} / 4. \quad (11.18)$$

The approach of this activation energy to this maximum is controlled by the $\mathbf{q}$ -dependences of the factors within the summation of Eq. (11.15).

The dependences of the non-adiabatic semiclassical jump rate's activation energy on jump distance are illustrated in Fig. 11.4 for long-range and short-range models of the electron–phonon interaction. In these examples, hops are taken to be between two equivalent electronic states of radius b separated by the distance h . For ionic materials' Fröhlich long-range electron–phonon interaction the activation energy given by Eq. (11.15) rises gently with increasing h , roughly as $(1/b - 1/h)$ at large separations. This behavior occurs because short hops in ionic materials require smaller shifts of distant ions than do long hops. By contrast, as depicted in Fig. 11.4, a short-range electron–phonon interaction generally produces a relatively steep rise of the energy of Eq. (11.15) with increasing h . The short-range interaction of Holstein's molecular-crystal model (MCM) produces the most extreme example (Holstein, 1959a). In particular, in the limit of dispersion-less optic phonons the MCM gives $E_A^{g,g'} = E_A^{\max} = F^2/4k$ for $h \neq 0$ because $f_g(\mathbf{q}) = f_{g'}(\mathbf{q}) = 1$. That is, this model gives the maximum activation energy applying to all non-adiabatic hops. This artifact results because the MCM implicitly presumes that the localized electronic states between which a carrier hops have arbitrarily small radii, $b \rightarrow 0$; see the discussion following Eq. (4.18).

As described by Eq. (11.14) and illustrated in Fig. 11.2, a small-polaron jump rate passes from an Arrhenius regime to a non-Arrhenius regime as the temperature is lowered below a significant fraction of the temperature corresponding to the maximum energy of the phonons with which a carrier principally interacts. Equations (11.5) and (11.6) indicate that the specific non-Arrhenius temperature dependence of the jump rate depends on (1) the range of the electron–phonon interaction, (2) the dispersion relations of the involved phonons, and (3) the separation between a hop's two sites. These non-Arrhenius temperature dependences often closely resemble

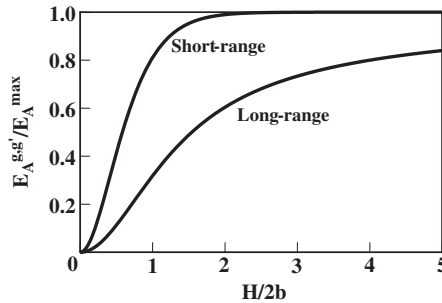


Fig. 11.4 The non-adiabatic hopping activation energy $E_A^{g,g'}$ in units of its maximum value $E_A^{\max}$ is plotted versus the jump length $h \equiv |\mathbf{g} - \mathbf{g}'|$ in units of twice the localized-state radius $2b$ for a hop between equivalent states $b_g = b_{g'} \equiv b$. The long-range electron–phonon interaction of an ionic medium produces a milder range-dependence than does a short-range electron–phonon interaction.

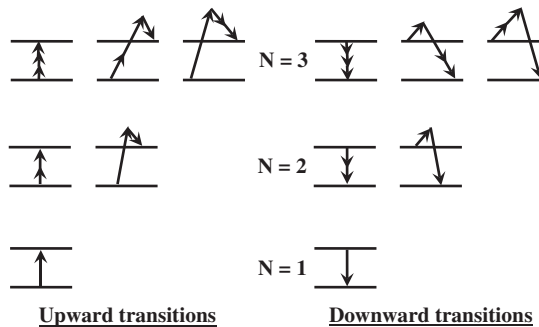


Fig. 11.5 Schematic description of processes in which the absorption and emission of one, two, and three acoustic phonons contribute to a transition upward or downward in energy.

those which are obtained by assuming percolative hopping with Arrhenius rates amongst energetically inequivalent sites (Emin, 1974).

Phonon-assisted transfers generally occur with multiple phonons being emitted and absorbed. Some processes involving the absorption of multiple acoustic phonons are illustrated in Fig. 11.5. In the extreme low-temperature limit when processes involving non-minimal absorption of phonon energy are completely frozen out, the processes depicted in the first columns of Fig. 11.5 persist. Then, since Eq. (11.6) gives $F_{g,g'}(\tau) \rightarrow 0$ as $T \rightarrow 0$, it is efficacious to expand $\exp[F_{g,g'}(\tau)]$ as a power series in the integrand of Eq. (11.4) to obtain:

$$R_{g,g'} = \left(\frac{t_{g,g'} e^{-S_{g,g'}}}{\hbar} \right)^2 e^{-\Delta_{g,g'}/2kT} \sum_{n=1}^{\infty} \frac{1}{n!} \int_{-\infty}^{\infty} d\tau [F_{g,g'}(\tau)]^n \cos(\Delta_{g,g'}\tau/\hbar). \quad (11.19)$$

Here n , the power of $F_{\mathbf{g},\mathbf{g}'}(\tau)$ in this series expansion, denotes the number of involved acoustic phonons and the τ -integration establishes energy conservation. That is, the net energy of the emitted or absorbed phonons equals $\Delta_{\mathbf{g},\mathbf{g}'}$.

Evaluating the integrals for the deformation-potential interaction in the long-hop limit, $h \equiv |\mathbf{g} - \mathbf{g}'| \rightarrow \infty$, with $|\Delta_{\mathbf{g},\mathbf{g}'}| < \hbar\omega_m$, where ω_m is the highest frequency acoustic phonon with which the localized carriers interact effectively, yields the low-temperature limit of the jump rate (Emin, 1974):

$$R_{\mathbf{g},\mathbf{g}'} = \frac{2\pi(t_{\mathbf{g},\mathbf{g}'}e^{-S_{\mathbf{g},\mathbf{g}'}})^2}{|\Delta_{\mathbf{g},\mathbf{g}'}|\hbar} e^{-(\Delta_{\mathbf{g},\mathbf{g}'} + |\Delta_{\mathbf{g},\mathbf{g}'}|)/2\kappa T} \sum_{n=1} \frac{[6\gamma(\Delta_{\mathbf{g},\mathbf{g}'}/\hbar\omega_D)^2]^n / (2n-1)!}{n!} \quad (11.20)$$

where $\gamma \equiv \gamma_{\text{DP}}(q_D)$ of Eq. (11.10). In this limit the jump rate for a hop downward in energy, $\Delta_{\mathbf{g},\mathbf{g}'} < 0$, becomes temperature independent while the jump rate for a hop upwards in energy, $\Delta_{\mathbf{g},\mathbf{g}'} > 0$, varies as $\exp(-\Delta_{\mathbf{g},\mathbf{g}'}/\kappa T)$. In addition to Eq. (11.20) the general expression Eq. (11.4) indicates that these phonon-assisted jump rates satisfy the detailed-balance condition, $R_{\mathbf{g},\mathbf{g}'}/R_{\mathbf{g}',\mathbf{g}} = \exp(-\Delta_{\mathbf{g},\mathbf{g}'}/\kappa T)$.

When $|\Delta_{\mathbf{g},\mathbf{g}'}| > \hbar\omega_m$, non-vanishing contributions to the summation begin at $n_m \equiv |\Delta_{\mathbf{g},\mathbf{g}'}|/\hbar\omega_m$, the minimum number of acoustic phonons required by energy conservation. When $2n - 1$ is large enough Stirling's approximation can be applied to its factorial which is then combined with the expansion parameter to yield $\sim 6\gamma(\Delta_{\mathbf{g},\mathbf{g}'}/n\hbar\omega_D)^2$. Analysis then shows that the downward jump rate falls once $|\Delta_{\mathbf{g},\mathbf{g}'}|$ rises above $\sim \gamma\hbar\omega_m$. Similarly, the high-temperature jump rate for a hop downward in energy, Eq. (11.17), falls once $|\Delta_{\mathbf{g},\mathbf{g}'}| > 4E_{\mathbf{g},\mathbf{g}'}$. This feature indicates a bottleneck; the disorder energy becomes so large that the electron-phonon interaction is no longer strong enough to enable the multi-phonon emission associated with a downward transition to occur rapidly. As a result hopping downward in energy is suppressed when driven by a strong enough electric field (Emin, 1975a). This effect produces negative differential conductivity for hopping conduction (Emin and Hart, 1985). The occurrence of such bottlenecks also defines the “inverted” regime of chemistry's electron transfer reactions (Marcus, 1960).

11.5 Semiclassical treatment of the non-adiabatic polaron jump rate

The high-temperature jump rate in Eq. (11.17) is semiclassical in that it can be derived by treating atomic vibrations as classical (Holstein, 1959b). In this approach the electronic energy of a carrier on a site is taken to be a function of atomic positions that are themselves explicit functions of time τ . The jump amplitude $a_{\mathbf{g}'}(\tau)$ for a

carrier that begins on site $\mathbf{g}$ and moves to site $\mathbf{g}'$ under the influence of a small electronic transfer energy $t_{\mathbf{g},\mathbf{g}'}$ is governed by the differential equation:

$$\left[i\hbar \frac{\partial}{\partial \tau} - E_{\mathbf{g}'}(\tau) \right] a_{\mathbf{g}'}(\tau) = t_{\mathbf{g},\mathbf{g}'} \exp \left[-i \int_0^\tau d\tau' E_{\mathbf{g}}(\tau') / \hbar \right], \quad (11.21)$$

where the exponential factor on the right-hand-side of Eq. (11.21) is the phase integral for a carrier confined to site $\mathbf{g}$. Solving the differential equation to first order in $t_{\mathbf{g},\mathbf{g}'}$ yields

$$a_{\mathbf{g}'}(\tau) = \exp \left[-i \int_0^\tau d\tau' E_{\mathbf{g}'}(\tau') / \hbar \right] \frac{t_{\mathbf{g},\mathbf{g}'}}{i\hbar} \int d\tau \exp \left\{ i \int_0^\tau d\tau' [E_{\mathbf{g}'}(\tau') - E_{\mathbf{g}}(\tau')] / \hbar \right\}. \quad (11.22)$$

The principal contributions to the second τ' -integration occur near points of stationary phase $\tau' = \tau_i$ defined by the coincidence condition $E_{\mathbf{g}}(\tau_i) = E_{\mathbf{g}'}(\tau_i)$. Near a point of stationary phase $E_{\mathbf{g}}(\tau') - E_{\mathbf{g}'}(\tau') \approx \dot{E}_{\mathbf{g},\mathbf{g}'}(\tau' - \tau_i)$ with $\dot{E}_{\mathbf{g},\mathbf{g}'}$ denoting the temporal rate of change of the energy difference $E_{\mathbf{g}} = E_{\mathbf{g}'}$ evaluated at $\tau' = \tau_i$. Using this approximation the second temporal integration is performed to obtain the transition amplitude associated with a particular coincidence event.

Most generally $a_{\mathbf{g}'}(\tau)$ comprises a series of contributions from different coincidence events. The jump probability is the absolute square of this sum of transition amplitudes. However correlations between transition amplitudes from different coincidence events are eliminated with sufficient three-dimensional vibratory dispersion (de Wit, 1968; Emin, 1975a). The net probability of a jump is then the sum of the absolute squares of the jump amplitudes which are each associated with a coincidence event. The probability of a hop in response to a coincidence event characterized by $t_{\mathbf{g},\mathbf{g}'}$ and $\dot{E}_{\mathbf{g},\mathbf{g}'}$ is (Holstein, 1959b; Emin, 1991a):

$$P(\dot{E}_{\mathbf{g},\mathbf{g}'}) = \frac{2\pi |t_{\mathbf{g},\mathbf{g}'}|^2}{\hbar |\dot{E}_{\mathbf{g},\mathbf{g}'}|}. \quad (11.23)$$

This probability measures the ability of a carrier to move between sites during the time window presented to it by a coincidence event. That is, P is essentially the square of the product of the rate with which the carrier can move between sites having near equal electronic energies $|t_{\mathbf{g},\mathbf{g}'}|/\hbar \equiv 1/\tau_i$ and the duration of its coincidence $\tau_c \equiv (\hbar/\dot{E}_{\mathbf{g},\mathbf{g}'})^{1/2}$. A coincidence's finite duration is a quantum-mechanical phenomenon: τ_c multiplied by the electronic-energy change during this time interval $|\dot{E}_{\mathbf{g},\mathbf{g}'}| \tau_c$ satisfies the Heisenberg uncertainty relation, $|\dot{E}_{\mathbf{g},\mathbf{g}'}| \tau_c^2 \sim \hbar$ (Emin, 1973b).

Integrating the product of $P(\dot{E}_{g,g'})$ and the rate with which the corresponding coincidences occur over all thermally accessible values of $\dot{E}_{g,g'}$ yields the semiclassical jump-rate expression in Eq. (11.17) (Holstein, 1959b; Emin, 1991a). Its activation energy is simply the minimum strain energy required to establish a coincidence (Friedman and Holstein, 1963; Emin, 1973b). Thus the activation energy of Eq. (11.17) vanishes if the energy of the equilibrated carrier at g happens to equal that of the state at site g' .

For a hop to be non-adiabatic $P \ll 1$ its electronic transfer energy must be small enough: $|t_{g,g'}|^2 \ll \hbar |\dot{E}_{g,g'}| / 2\pi \approx \hbar \omega (E_A \kappa T)^{1/2}$. Furthermore the semiclassical treatment requires $\hbar \omega (E_A)^{1/2} \ll (\kappa T)^{3/2}$ to assure that atoms' velocities do not change appreciably during a coincidence (Holstein, 1959b). Thus, a semiclassical non-adiabatic hop necessitates $|t_{g,g'}| \ll \kappa T$ (Emin and Holstein, 1969). The smallness of the electronic transfer energy implies a correspondingly small value for the pre-exponential factor in the expression for a non-adiabatic semiclassical polaron hop, Eq. (11.17). Thus, non-adiabatic semiclassical small-polaron hopping is characterized by very small values of the pre-exponential factors of its Arrhenius mobility and dc conductivity, $\ll 0.1 \text{ cm}^2/\text{V-sec}$ and $\ll 10^2/\Omega\text{-cm}$ at 300 K, respectively. Observation of such small values might also indicate non-adiabatic hopping between defects and impurities since these hops are especially long with exceptionally small electronic transfer energies.

11.6 Semiclassical treatment of the adiabatic polaron jump rate

The rate for a phonon-assisted hop is generally larger in the adiabatic regime, where the electronic carrier moves rapidly enough to adjust to relevant atomic movement, than in the non-adiabatic regime, where the carrier is unable to follow such atomic motion. Nonetheless adiabatic and non-adiabatic hopping both manifest similar temperature dependences. As depicted in Fig. 11.2 low-temperature non-Arrhenius thermally assisted jump rates generally evolve to Arrhenius temperature dependences as the temperature is raised high enough for atoms' vibrations to be treated as classical. A hop in this Arrhenius semiclassical regime requires atomic displacements that bring the electronic energies of initial and final states into coincidence with one another. In other words, a semiclassical adiabatic hop requires establishing an atomic configuration for which the probabilities of a carrier occupying a hop's initial and final sites equal one another.

For the molecular-crystal model, adiabatic and non-adiabatic hops from site 1 to site 2 can be described with the aid of the anharmonic adiabatic potentials depicted in Fig. 11.6. Occupation of the left-hand well of lowest adiabatic potential corresponds to the carrier occupying site 1 while occupation of this adiabatic potential's right-hand well corresponds to the carrier occupying site 2. A semiclassical hop

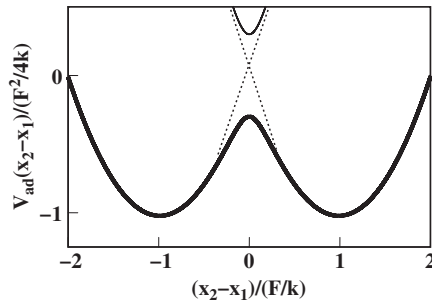


Fig. 11.6 The thick and thin solid curves are the lower and upper two-site adiabatic potentials of Eq. (3.14) plotted in units of $F^2/4k$ against the relative atomic displacement coordinate $x_2 - x_1$ in units of F/k for $F^2/4k = 3t$. Transitions between adiabatic potentials are depicted with dashed lines. The energy difference between either of the thick curve's minima and its maximum at $x_1 = x_2$ is the hopping activation energy $F^2/4k - t$. The lower adiabatic potential is strongly anharmonic in the vicinity of its maximum.

between sites 1 and 2 is described by trajectories which take a fictitious particle between the two wells of the lowest adiabatic potential. At a coincidence a trajectory can either remain on its adiabatic potential or make a non-adiabatic transition between adiabatic potentials. The trajectory for an adiabatic hop remains on the lowest adiabatic potential as it passes between its two wells. A trajectory for a non-adiabatic hop (1) transitions from the lower adiabatic potential to the upper adiabatic potential, (2) makes an odd number of oscillations through the upper adiabatic potential's minimum, and then (3) transitions from the upper adiabatic potential to the lower adiabatic potential. Even in the limit where non-adiabatic transitions predominate, a hop requires an adiabatic passage on either the upper or lower adiabatic potential. The non-adiabatic jump rate calculated in the preceding section is the rate for such an adiabatic passage when the electronic-transfer energy is arbitrarily small.

The net probability of a carrier transferring from site 1 to site 2, $W_t(1,2)$, is found by summing the probabilities of the direct adiabatic trajectory and all those involving non-adiabatic transitions (Holstein, 1959b):

$$W_t(1,2) = 2 \frac{1 - e^{-(t_{1,2}/t_m)^2}}{2 - e^{-(t_{1,2}/t_m)^2}}, \quad (11.24)$$

where $t_m \equiv (\hbar | \dot{E}_{1,2} | / 2\pi)^{1/2}$. Here $\exp[-(t_{1,2}/t_m)^2]$ is the probability of a non-adiabatic transition near the coincidence when its electronic transfer energy is $t_{1,2}$ (Zener, 1932). The plot of $W_t(1,2)$ versus $t_{1,2}/t_m$ in Fig. 11.7 shows that the non-adiabatic regime, where $W_t(1,2)$ is proportional to the square of $t_{1,2}$, is restricted to extremely

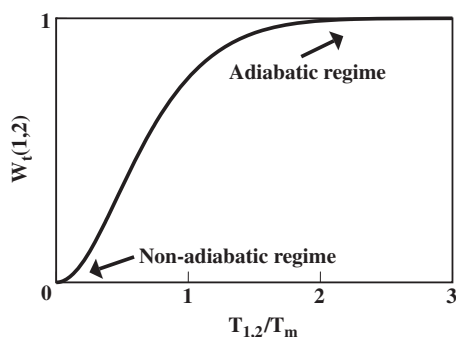


Fig. 11.7 The probability of a carrier moving from site 1 to site 2 in response to a coincidence increases with the electronic transfer energy $t_{1,2}$. Non-adiabatic hops, characterized by this transfer probability being proportional to the square of $t_{1,2}$, require extremely small transfer energies: $t_{1,2} \ll t_m$, where $t_m^2 \approx \hbar\omega(E_A kT)^{1/2}$. The adiabatic regime is defined by this transfer probability approaching unity.

small values of the electronic transfer energy, $t_{1,2} \ll t_m$. The adiabatic regime, where $W_t(1,2)$ approaches unity, occurs for $t_{1,2} > t_m$. Estimates of nearest-neighbor electronic transfer energies frequently find them large enough, $> t_m \sim (\hbar\omega)^{1/2}(E_A kT)^{1/4}$, for polaron hopping to be in the adiabatic regime (Herring, 1961). Nonetheless polaron hopping is often assumed to be in the non-adiabatic regime.

In the adiabatic limit, where a carrier always transfers between sites upon experiencing a coincidence, the potential governing jump-related atomic vibrations, the bold curve of Fig. 11.6, is anharmonic. By contrast, in the non-adiabatic limit, where a carrier rarely transfers between sites upon experiencing a coincidence, the potential governing jump-related atomic vibrations, the bold curve of Fig. 11.3, is harmonic.

Two complementary classes of semiclassical adiabatic polaron hop have been considered. These two types of adiabatic hop are represented by different kinds of trajectory that pass between the two minima of the lower anharmonic adiabatic potential of Fig. 11.6. In the weak-dispersion regime the amplitude of a trajectory that initially vibrates about one minimum gradually expands to transcend the energy barrier and encompass both minima. A trajectory that repeatedly transcends both wells and the negative-curvature region between them has its average frequency lowered (Emin, 2008). Ultimately after repeated passes the amplitudes of these double-well oscillations slowly diminish and the trajectory settles into oscillations about the other minimum. All told, the weak-dispersion regime envisions a hop's trajectory being slowly modulated by the transfer of energy between those atomic vibrations coupled to the carrier and other atomic vibrations. For example, energy transfer between vibrations of the molecules of Holstein's molecular-crystal model generates dispersion of the model's optic-vibration modes (Holstein, 1959b).

An adiabatic semiclassical hop in the strong-dispersion regime is represented by a ballistic-type of trajectory. Its trajectory suddenly rises from oscillations near the bottom of a potential well to a position above the energy barrier and then rapidly descends to oscillate about the double-well's other minimum. The hop is presumed to occur with no immediate return (Holstein, 1959b; Emin and Holstein, 1969). This presumption can be justified if a sufficiently large number of vibration modes having very disparate frequencies are strongly coupled to the carrier. Then phase coherence between these vibrations is lost within a typical vibration period. Thus interference obliterates the longer-lasting coherence characteristic of the weak-dispersion regime.

Several features of a semiclassical adiabatic hop enhance its rate above that of a non-adiabatic hop. First, as depicted in Fig. 11.7, the probability that a carrier will hop in response to a coincidence event is much larger for adiabatic hopping than for non-adiabatic hopping (Holstein, 1959b). Second, as illustrated in Fig. 11.6, transfer of a carrier in response to atomic movement lowers the peak of the adiabatic potential thereby reducing the hop's activation energy (Emin and Holstein, 1969). Third, in the weak-dispersion regime the reduction of the average vibration frequency for trans-barrier oscillations lowers these vibrations' free-energy. This effect augments the rate of slowly modulated coincidence formation (Emin, 2008).

Here the jump rate for an adiabatic semiclassical hop between degenerate sites is calculated for the weak-dispersion regime. The jump rate in the strong-dispersion regime is then obtained simply by neglecting the contribution to the weak-dispersion regime's adiabatic rate that comes from slowly modulated atomic oscillations. This result agrees with that obtained by directly calculating the rate for a semiclassical adiabatic hop in the strong-dispersion regime (Emin and Holstein, 1969).

In the adiabatic limit a carrier always follows atoms' motion. Concomitantly, as described in Chapter 4, these atoms' motions are also affected by a carrier's response to them. A polaron's adiabatic hopping rate is governed by the resulting atomic motion. At sufficiently high temperatures atoms' motion may be treated as classical. Then a hop requires atoms to pass through a coincidence configuration where the probabilities of the carrier occupying the jump's initial and final sites are equal. The semiclassical adiabatic jump rate generally has the form (Emin, 2008):

$$R_{\text{ad}} = \nu \exp[-(F_c - F_i)/\kappa T], \quad (11.25)$$

where ν is a representative atomic vibration frequency. Here F_c represents the system's free energy with atoms vibrating about the positions they assume during the hop's minimum-energy coincidence event. A coincidence is indicated by the $x_2 - x_1 = 0$ peak in Fig. 11.6 but appears as a saddle-point of the adiabatic potential

when plotted against multiple deformation coordinates. The system's free energy with atoms vibrating about the displaced equilibrium positions they assume when the carrier occupies the hop's initial site is denoted by F_i .

Confining the self-trapped carrier at the hop's initial site shifts the equilibrium positions about which surrounding atoms vibrate. These atomic positions and associated stiffness constants are determined by minimizing the system's energy with respect to atomic displacements. The associated free energy F_i is the sum of energies arising from shifting atoms' equilibrium positions and the free energy of atoms' vibrations about these equilibrium positions:

$$F_i = \left(\frac{T_e}{R_i^2} - \frac{V_S}{R_i^3} - \frac{V_L}{R_i} \right) + f_i. \quad (11.26)$$

The lowering of the system's energy produced by shifts of atoms' equilibrium positions, the bracketed term of Eq. (11.26), depends on the parameters governing the formation of the self-trapped state of dimensionless radius R_i . These parameters, the carrier's kinetic energy T_e , its short-range electron–phonon interaction V_S and its long-range electron–phonon interaction V_L are defined in Eqs. (4.31)–(4.33). The self-trapped carrier's radius R_i minimizes the bracketed term of Eq. (11.26) subject to the structural constraints of a real solid. For example, the self-trapped carrier cannot be smaller than the relevant single structural unit: a molecule, one of its subunits, a bond or an ion (Emin, 2000a). The free energy associated with atoms harmonically vibrating about these displaced equilibrium positions is denoted by f_i .

At a coincidence the carrier is shared equally between initial and final sites (Emin, 2000a). A hop's characteristic coincidence is that which minimizes the coincidence energy with respect to atomic displacements (Friedman and Holstein, 1963). The free energy associated with atoms vibrating about the positions they assume at the minimum-energy coincidence is:

$$F_c = \left[\frac{T_e}{R_c^2} - \frac{V_S}{2R_c^3} - \frac{V_L}{2} \left(\frac{1}{R_c} + \frac{1}{S} \right) \right] - |t(S)| + f_c, \quad (11.27)$$

where the dimensionless separation between sites S greatly exceeds R_c , the dimensionless radius of each half of the shared self-trapped carrier (Emin, 2000a). Here the radius R_c minimizes the square-bracketed term of Eq. (11.27) subject to the solid's structural constraints (Emin, 2000a). The numerical coefficients of contributions to F_c that are explicitly related to electron–phonon interactions are half those of comparable terms in Eq. (11.26). This feature, reflective of the nonlinearity of self-trapping phenomena, occurs because dividing the self-trapped carrier into two equal parts reduces the polaron-related energy of each half by four: $2(1/4) = 1/2$. The

electronic carrier's expansion to the hop's two sites also lowers the carrier's confinement energy by $t(S)$, the electronic transfer energy linking the two sites. Finally, f_c denotes the free energy of atomic vibrations associated with the coincidence configuration.

Combining Eqs. (11.26) and (11.27) with Eq. (11.25) yields:

$$R_{ad} = \nu \exp[(f_i - f_c)/\kappa T] \exp(-E_a/\kappa T), \quad (11.28)$$

where the activation energy is

$$E_a \equiv T_e \left(\frac{1}{R_c^2} - \frac{1}{R_i^2} \right) - V_s \left(\frac{1}{2R_c^3} - \frac{1}{R_i^3} \right) - V_L \left(\frac{1}{2R_c} + \frac{1}{2S} - \frac{1}{R_i} \right) - |t(S)|. \quad (11.29)$$

Generally $R_c \geq R_i$ since the R_c -dependent electron-phonon interaction terms of Eq. (11.27) are smaller than the analogous terms of Eq. (11.26) (Emin, 2000a). Nonetheless, a common simplification is to set $R_c = R_i \equiv R$. Then the semiclassical adiabatic hopping activation energy becomes

$$E_a = \frac{V_s}{2R^3} + \frac{V_L}{2} \left(\frac{1}{R} - \frac{1}{S} \right) - |t(S)|. \quad (11.30)$$

This activation energy generally rises with increasing inter-site separation. By itself, this effect fosters a carrier executing short hops.

Movement of an electronic carrier in response to atoms' vibrations generally reduces the associated stiffness constants thereby decreasing the vibrations' free energy (Emin, 2000a). This effect is especially large for a carrier's semiclassical adiabatic hop (Emin, 2008). Near a coincidence saddle-point the carrier transfers between sites with their atomic deformation energies changing by only the jump's electronic transfer energy. As a result, the carrier-induced lowering of vibrations' free energy which enters into Eq. (11.28) is dominated by that attributed to a coincidence.

The lowering of vibrations' free energy arising from a coincidence is addressed by considering matrix elements of the electron-phonon interaction between the closely spaced pair of electronic states associated with the coincidence. In particular, evaluation of Eq. (4.37) for the even-symmetry lowest adiabatic electronic wavefunction $\Psi_+(\mathbf{r})$ and its odd-symmetry first excited state $\Psi_-(\mathbf{r})$ yields

$$\frac{f_i - f_c}{\kappa T} \cong \frac{\int d\mathbf{r} \Psi_+^*(\mathbf{r}) \Psi_-(\mathbf{r}) \int d\mathbf{r}' \Psi_+(\mathbf{r}') \Psi_-^*(\mathbf{r}') I(\mathbf{r}, \mathbf{r}')}{t_{1,2}}. \quad (11.31)$$

Here $I(\mathbf{r}, \mathbf{r}')$ is the electron–phonon interaction function defined in Eq. (4.23) and the two adiabatic electronic wavefunctions are evaluated at the coincidence between sites 1 and 2 where their energies are separated by $2t_{1,2}$.

These two adiabatic electronic wavefunctions are then represented in terms of local electronic functions, $\phi_1(\mathbf{r})$ and $\phi_2(\mathbf{r})$, that, respectively, represent a carrier occupying sites 1 and 2:

$$\Psi_{\pm}(\mathbf{r}) \equiv \frac{\phi_1(\mathbf{r}) \pm \phi_2(\mathbf{r})}{\sqrt{2}}. \quad (11.32)$$

Writing $I(\mathbf{r}, \mathbf{r}')$ as the sum of its long-range and short-range components, Eqs. (4.27) and (4.28), and taking $\phi_1(\mathbf{r})$ and $\phi_2(\mathbf{r})$ as having identical forms albeit displaced from one another by the inter-site separation, Eq. (11.31) is evaluated to yield (Emin, 2008):

$$\frac{\Delta f_{\text{vib}}^c}{\kappa T} = \frac{E_a + t}{t}. \quad (11.33)$$

Feynman provides an alternative method of calculating the shift of vibrations' free energy produced by carrier-induced anharmonicity (Feynman, 1972; Emin, 2012).

Incorporating Eq. (11.33) into Eq. (11.28) yields the semiclassical adiabatic jump rate for the weak-dispersion model:

$$R_{\text{ad}} \cong v \left[\exp\left(\frac{E_a}{t}\right) \right] \exp\left(-\frac{E_a}{\kappa T}\right), \quad (11.34)$$

where $E_a \gg t \gg t_m \approx (\hbar\omega)^{1/2}(E_A\kappa T)^{1/4}$. The square-bracketed factor in Eq. (11.34) rises as the activation energy E_a is increased. This effect tends to offset the fall of the thermally activated factor with increasing activation energy. This feature is akin to the empirically observed Meyer–Neldel compensation effect (Meyer and Neldel, 1937).

The adiabatic jump rate in the strong-dispersion regime has been computed with the occurrence-probability approach. The occurrence-probability approach for ballistic trajectories ignores atoms' vibrations and identifies the adiabatic jump rate with the thermally averaged rate of coincidence formation with $R_{\text{ad}} \equiv \langle \dot{E}_{1,2} \rangle \langle \delta(E_1 - E_2) \rangle$, where the first average is over atomic velocities and the second average is over atomic positions (Holstein, 1959b; Emin and Holstein, 1969). The resulting rate is given by Eq. (11.28) but with $f_i - f_c = 0$.

Adiabatic hopping is distinguished from non-adiabatic hopping by the magnitude of the non-Arrhenius factor of its jump rate. In particular, in the weak-dispersion

regime the square-bracketed factor of the adiabatic rate, Eq. (11.34), is larger than one. In the strong-dispersion regime the square-bracketed term of the adiabatic rate is just unity. By comparison the corresponding square-bracketed factor of the non-adiabatic rate is always very much less than one:

$$R_{\text{non-ad}} = v \left[\left(\frac{t}{t_m} \right)^2 \right] e^{-E_A/\kappa T} = \left(\frac{t^2}{\hbar} \right) \sqrt{\frac{\pi}{4\kappa T E_A}} e^{-E_A/\kappa T}, \quad (11.35)$$

where $t \ll t_m$ and Eq. (11.17) has been rewritten with $\Delta_{g,g'} = 0$. Thus the magnitude of a jump rate's non-activated factor indicates whether the hop is adiabatic or non-adiabatic.

Several effects contribute to the small-polaron jump rate usually falling with increasing inter-site separation. First, the hopping activation energies generally rise with increasing inter-site separation. Even with the Meyer–Neldel compensation effect of Eq. (11.34) the adiabatic limit's jump rate falls with increasing inter-site separation provided that $t > \kappa T$. Second, as depicted in Fig. 11.7, the decrease of electronic transfer energies with increasing inter-site separation causes the jump rate to fall out of the adiabatic regime. Inter-site electronic transfer then becomes too sluggish for carriers to always avail themselves of the opportunities to hop presented by atoms being displaced into coincidence configurations. At large enough inter-site separations the electronic transfer energy becomes so small, $t \ll t_m \sim (\hbar\omega)^{1/2} (E_A\kappa T)^{1/4}$, that the jump rate falls into its non-adiabatic regime where it is proportional to the square of the electron-transfer energy (Holstein, 1959b).

Applying an electric field of magnitude F adds a contribution to the hopping activation energy. The added contribution lowers the activation energy for jumps facilitated by the field and raises the hopping activation energy for jumps in the opposite direction. In sufficiently strong fields the current flow becomes non-Ohmic as carriers overwhelmingly jump in concert with the applied field (Emin, 1975a).

The electric-field-dependent change of the potential energy of a carrier as it hops is typically smaller than the field-free activation energy. Then the electric field only contributes an additive correction to the hopping activation energy. This contribution is half the change of the carrier's potential energy garnered in the hop (Emin, 1975a). Thus the field-dependent contribution to a hop's activation energy is proportional to the jump's length.

As noted below Eq. (11.30), field-independent contributions to the activation energy also generally increase with a hop's length. With the long-range electron–phonon interaction of polar and ionic materials the jump-distance-dependent contribution to the hopping activation energy increases in proportion to the negative of the inverse jump distance. Such a hop's activation energy rises with increasing jump

distance because greater shifting of distant ions is then required to produce a coincidence.

These two effects combine to contribute an explicit jump-length-dependent factor to the jump rate for a hop in concert with the applied electric field F (Emin, 2008):

$$\exp \left[\left(\frac{eFaS}{2} + \frac{V_L}{2S} \right) / \kappa T \right] = \exp \left\{ \frac{\sqrt{eFaV_L}}{\kappa T} \left[1 + \frac{(S-S_0)^2}{2SS_0} \right] \right\}. \quad (11.36)$$

Here S is the jump distance in units of the interatomic separation a , the term containing V_L comes from Eq. (11.30), and $S_0 \equiv (V_L/eFa)^{1/2} = [e(1/\epsilon_\infty - 1/\epsilon_0)/2F]^{1/2}/a$. Typically $S_0 \sim 10/a(\text{\AA})$ for $F = 10^6$ V/cm, where $a(\text{\AA})$ is the interatomic separation measured in angstroms.

Longer-range hops whose rates are suppressed by their relatively large field-free activation energies are enabled by the application of a sufficiently strong electric field. The factor in Eq. (11.36) becomes especially simple when the electric field's strength is large enough to lower S_0 to close to the suppressed hops' characteristic jump distance S : $(S - S_0)^2 < 2S_0^2$. The electric-field dependence of the jump rate then rises with electric-field strength as $\exp[CF^{1/2}/\kappa T]$ with $C \equiv [e^3(1/\epsilon_\infty - 1/\epsilon_0)/2]^{1/2}$. Measures of electric-field driven charge transport possessing this form are said to manifest Frenkel–Poole behavior (Frenkel, 1938).

11.7 Vibrational relaxation and correlations between polaron jumps

The transfer of energy between vibrating atoms is essential to phonon-assisted hopping. For a thermally activated semiclassical hop to occur, the atoms involved in forming a coincidence must absorb an extraordinary amount of energy from the vibrations of surrounding atoms. After the hop's charge transfer this vibrational energy is dissipated to surrounding atoms.

The transfer of energy between harmonically vibrating atoms is governed by the dispersion of their vibration frequencies. Holstein's molecular-crystal model envisions the carrier to be coupled to a single band of optic phonons (Holstein, 1959b). As such, it generally takes much longer than a typical optic-phonon vibration period $1/\omega$ to transfer a significant fraction of the strain energy utilized in forming a coincidence. In particular, the vibration energy remaining with atoms involved in a coincidence event produced with optic phonons after the time τ is $\sim E_A(\omega_w\tau)^{-d/2}$ for $\omega_w\tau > 1$, where ω_w is the width of the band of optic phonons and d is the vibrational system's dimensionality (Emin, 1970; 1971a; 1971b). The vibrational "hot spot" associated with a coincidence is deemed to have relaxed when its residual vibrational energy falls to κT . With $E_A \gg \kappa T$ and $\omega \gg \omega_w$ this relaxation time is much greater than $1/\omega$.

During relaxation from a coincidence the rates with which the involved sites form additional coincidences are enhanced. During a coincidence's relaxation there is also an increased probability that one of its sites will form a coincidence with a site that was not involved in the prior coincidence. In these instances the amount of energy that must be garnered from the vibrations of surrounding atoms to establish another coincidence rises as the energy utilized in forming the prior coincidence dissipates.

A carrier which responds adiabatically to two sites' recurring coincidences will hop back and forth between them so as to occupy the site of lowest electronic energy (Emin and Kriman, 1986; Emin, 1987). This situation is represented by trajectories that oscillate over the barrier of the adiabatic potential of Fig. 11.6. Furthermore a carrier that responds adiabatically to a coincidence with a fresh site will manifest an augmented jump rate prior to vibrational relaxation. These enhanced diffusion effects vanish in the extreme non-adiabatic limit where electronic transfer rates are much slower than coincidences' relaxation rates. A carrier's response is then too slow for it to benefit from the enhanced probability of successive coincidence events prior to vibrational relaxation.

When vibrational relaxation produces significant temporal correlations between the occurrences of successive coincidences a carrier's hops occur in flurries (Emin and Kriman, 1986; Emin, 1987). There are bursts of rapid hopping separated by long quiescent intervals. During a period of active hopping there are frequent return hops. A carrier's actual diffusion can occur when it abandons shuttling between two sites to begin back-and-forth motion between one of these sites and a fresh site. When in thermal equilibrium, active hopping and dormant periods compensate one another to yield the previously determined jump rates (Holstein, 1959b; Emin and Holstein, 1969; Emin, 2008). A series of Monte Carlo molecular dynamics calculations of the hopping diffusion of a vacancy in model substances report analogous correlation effects (Bennett and Alder, 1968; 1971; Bennett, 1975).

Polarons formed from carriers injected into a material do not begin in thermal equilibrium. Equilibration as polarons requires carriers' self-trapping with their stabilization energy being dissipated via vibrations' dispersion. Prior to their equilibration as polarons injected carriers may move with greatly enhanced mobility. Relaxation can be so slow that transient high-field hopping conduction can be qualitatively different from polarons' steady-state hopping (Emin and Kriman, 1986).

Imagine for example a charge carrier injected into a narrow-band insulator alighting on a molecule of Holstein's molecular-crystal model. As depicted in the left-hand column of Fig. 11.8, the carrier's arrival shifts the equilibrium positions of the molecule's atoms. As the occupied molecule relaxes to its shifted equilibrium

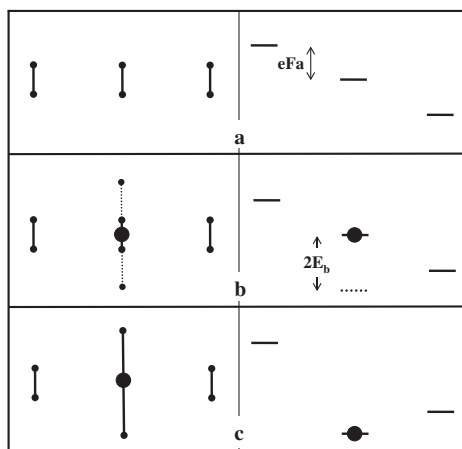


Fig. 11.8 Deformations and electronic energy levels are illustrated for three molecules of Holstein's molecular-crystal model with lattice constant a placed in electric field F . Panels *a* depict the carrier-free molecules at equilibrium. Panels *b* show the shift of the equilibrium deformation of a molecule (dashed line) and the corresponding lowering of its electronic energy $2E_b$ (dashed level) for $2E_b > eFa$ immediately after a carrier (the large dot) is placed on it. Panels *c* depict the equilibrated situation.

position vibrational energy comparable to Holstein's small-polaron binding energy E_b is transferred to surrounding molecules. The right-hand column of Fig. 11.8 indicates that the carrier's electronic state is not initially coincident with those of other molecules residing in the direction of an applied electric field F . However, the carrier's electronic energy passes through coincidences with molecules that reside in the down-field direction as the occupied molecule relaxes. If the carrier has a high probability of transferring in response to such a coincidence it will produce activation-less hopping with its mobility comparable to $ea^2v/\kappa T$, typically about $1 \text{ cm}^2/\text{V-sec}$ (Emin and Krivan, 1986; Emin, 1987). This relatively rapid transient hopping motion will cease once a carrier lingers long enough at a site to permit lattice relaxation.

The mobility determined in a time-of-flight mobility measurement depends on the relationship between the experiment's transit time and the carriers' relaxation times. Transit-time measurements performed on thin films may be short enough to reveal only the carriers' unrelaxed mobility (Emin and Krivan, 1986). Even carriers' transits through bulk materials may not afford sufficient time for carriers' full equilibration. For example, the multiple captures and releases of carriers from deep traps required for complete equilibration may take longer than carriers' transits through bulk materials (Gibbons and Spear, 1966).

11.8 Pair-breaking in semiclassical singlet-bipolaron hopping

Singlet small-bipolaron formation (cf. Section 7.2) is often described with the Holstein–Hubbard model. Then two carriers with opposing spins that share a common site displace the equilibrium value of its atomic-deformation configuration coordinate by twice that as from a single localized carrier. Since each of the two carriers benefits from the energy lowering produced by the doubled atomic displacements the pair’s net energy lowering is four times that for a solitary localized carrier $-E_b$. With U denoting the singlet pair’s mutual Coulomb repulsion, the singlet bipolaron’s net energy becomes $-4E_b + U$. Singlet-bipolaron formation is energetically favorable if this energy is less than the net polaron binding energy for two carriers localized at equivalent separated sites $-2E_b$. The small-bipolaron binding energy, the difference between these two energies, is $\varepsilon_b \equiv 2E_b - U$.

The four processes by which a singlet bipolaron can participate in charge transport are depicted in Fig. 11.9 (Emin, 1996). A bipolaron’s two carriers can hop in unison, one of a bipolaron’s two carriers can hop to another site, two polarons can merge to form a bipolaron, and a carrier can transfer from a bipolaron to a polaron.

The activation energies characterizing these types of semiclassical adiabatic hops have been calculated for the Holstein–Hubbard model (Emin, 1996). The activation energy for the joint hopping of the bipolaron’s two carriers is $E_A(\text{joint}) \equiv 2E_b - t_2$, where the first contribution is simply four times the corresponding contribution for a polaron hop and t_2 is the magnitude of the two-electron transfer energy associated with moving a pair of electronic carriers. The activation energy for the bipolaron’s separation into two polarons is $E_A(\text{separate}) \equiv (4E_b - U)^2/8E_b - t_1$, where t_1 is the magnitude of the conventional one-electron transfer energy and $t_1 > t_2$. The activation energy for uniting two polarons to form a bipolaron is $E_A(\text{unite}) \equiv U^2/8E_b - t_1$. As required by the detailed-balance condition the difference between the activation energies for separating a bipolaron into two polarons and that for merging two polarons into a bipolaron is just the bipolaron binding energy ε_b . Finally, the

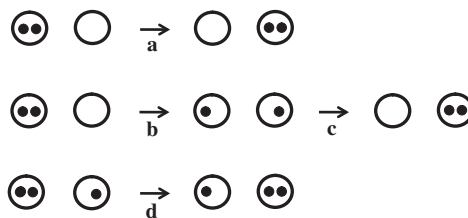


Fig. 11.9 Four transport processes involving a singlet bipolaron are (a) the joint transfer of a bipolaron’s two carriers, (b) the transfer of one of a bipolaron’s two carriers, (c) the uniting of two polarons into a bipolaron, and (d) the transfer of a carrier from a bipolaron to a polaron.

activation energy for transferring one of a bipolaron's two carriers to a polaron is found to be simply the activation energy for a single polaron hop, $E_A(\text{single}) \equiv E_b/2 - t_1$. With energetically stable bipolaron formation $\varepsilon_b > 0$, the activation energies for single-carrier transfers are smaller than that for the joint hopping of a bipolaron's two carriers: $E_A(\text{joint}) > E_A(\text{separate}) > E_A(\text{single}) > E_A(\text{unite})$.

Semiclassical dc hopping conductivity proceeds via the inter-site charge-transfer processes described above. The corresponding rates for these processes are: $R(\text{joint})$, $R(\text{separate})$, $R(\text{single})$ and $R(\text{unite})$. The dc conductivity is then (Emin, 1996):

$$\sigma = \frac{Ne^2a^2}{\kappa T} \{4f_b f_v [R(\text{joint}) + R(\text{separate})] + 2f_p(1-f_p)R(\text{single})\}, \quad (11.37)$$

where N is the site density and a is the nearest-neighbor separation. Here f_b and f_p respectively denote the fractions of sites occupied by bipolarons and polarons and f_v denotes the fraction of sites remaining vacant with $f_b + f_p + f_v = 1$. The numerical factors in Eq. (11.37) result because (1) $R(\text{joint})$ transports two charges rather than one, (2) each polaron state has a two-fold spin degeneracy, and (3) the contribution associated with merging polarons into bipolarons has been combined with that associated with separating bipolarons into polarons by using the detailed-balance relation $f_p f_p R(\text{unite}) = f_b f_v R(\text{separate})$.

The general expression for the dc conductivity of Eq. (11.37) can be culled to its most significant contribution. First, it is noted that $R(\text{separate}) > R(\text{joint})$ because $E_A(\text{separate}) < E_A(\text{joint})$. In addition, $R(\text{joint})$ will be reduced further if the two-electron transfer energy t_2 is small enough so that joint hops become non-adiabatic. Second, it is noted that $2f_p R(\text{single}) \gg 4f_b f_v R(\text{separate})$ in the physically relevant situation when most carriers form bipolarons: $f_b \approx f/2$, $f_v \approx 1 - f/2$ and $f_p \approx f^{1/2} \exp(-\varepsilon_b/2\kappa T)$, where f is the net number of charge carriers per site. In particular, the net activation energy for single-polaron hops, $\varepsilon_b/2 + E_A(\text{single})$, is considerably smaller than that for bipolaron's break-up, $E_A(\text{separate})$. Thus, the dc conductivity in the thermally activated semiclassical regime is dominated by polaron hops (Emin, 1996):

$$\sigma \cong \frac{2Ne^2a^2}{\kappa T} f_p R(\text{single}), \quad (11.38)$$

where the net activation energy is $3E_b/2 - U/2 - t_1$.

All told, polaron hopping becomes ever more predominant as E_b and the bipolaron binding energy ε_b are increased. In other words, as ever stronger binding increases the fraction of carriers that form bipolarons their hopping becomes so slow

that the semiclassical dc conductivity is dominated by the relatively rapid motion of the fraction of carriers that remain as polarons.

11.9 Exchange interactions and polaron hopping in magnetic semiconductors

Polaron hopping has been reported to occur in numerous magnetic semiconductors. In these instances a polaron's inter-site motion is affected by interactions of its self-trapped carrier with the electrons that generate the material's local magnetic moments. The local magnetic moments are solely attributed to electrons' spins when, as is common, orbital magnetic moments are quenched by a material's crystal field. Then exchange interactions among valence electrons having unpaired spins govern the host material's magnetic interactions. A polaron's self-trapped carrier experiences the material's magnetism through its exchange interactions with valence electrons having unpaired spins.

When obeying Hund's rule the spin of a carrier localized at a site aligns with its valence electrons' spins to produce a net spin state of maximum multiplicity. The transfer of a carrier between two sites then depends on the relative alignment of these sites' magnetic moments. In the classical limit, in which the local moments are presumed to be arbitrarily large, the carrier's transfer energy becomes proportional to the spinor overlap factor: $\cos(\theta/2)$, where θ is the angle between the two sites' local moments. Then $\cos(\theta/2) = 1$ for parallel alignment $\theta = 0$ and $\cos(\theta/2) = 0$ for anti-parallel alignment $\theta = \pi$. Thus, the classical limit prohibits hops between sites whose magnetic moments are anti-parallel. However, this simple classical limit can be misleading. In particular, for finite-sized local moments the factor corresponding to $\cos(\theta/2)$ with $\theta = \pi$ becomes $1/(2S + 1)$, where S is the net spin of a site's valence electrons (Anderson and Hasegawa, 1955). Consider for example an electron hopping in a $S = 1/2$ magnet or a hole hopping in a $S = 1$ magnet. Then, rather than being reduced to zero, the spinor overlap factor only falls to $1/2$ for electron transfer between sites with oppositely aligned $S = 1/2$ valence spins. The lowering of the transfer energy arising from such a modest reduction of spinor overlap generally will not be sufficient to qualitatively alter a polaron's hopping rate.

The above discussion has modeled the transfer of an electron in a magnetic material as a one-electron process constrained by intra-site exchange interactions. However, exchange interactions result from indistinguishable electrons' mutual Coulomb repulsions. Thus electron transfer between magnetic sites is treatable as a multi-electron process in which one electron moves from a site containing $n + 1$ unpaired indistinguishable electrons to a site containing n unpaired indistinguishable electrons (Liu and Emin, 1979; Emin and Liu, 1983). The simplest example emerges when addressing the transfer of an electron from a site containing two

electrons to a site containing one electron. For transitions between Hund's-rule states this approach regains the previously discussed spinor overlap factors. The multi-electron treatment also obtains overlap factors for transitions involving non-Hund's rule states. However, the transfer energy associated with transitions among Hund's rule states differs from its one-electron value in that it contains direct and exchange contributions involving Coulomb interactions between the carrier and valence electrons. Different combinations of these two-electron energies contribute to the transfer energies associated with non-Hund's rule states.

Magnetic ordering generally affects the activation energy for semiclassical polaron hopping. The most significant effect occurs with antiferromagnetic ordering. The activation energy for hopping between different sublattices of an antiferromagnet increases by about κT_N upon lowering the temperature below the Néel temperature T_N (Liu and Emin, 1979; Emin and Liu, 1983). The activation energy is augmented for these polaron hops because they occur between sites for which the ordered-state's internal magnetic field changes sign.